BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, RAWALPINDI

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

Mathematics (Subjective) — (For All Sessions) — HSSC-(P-II)-A/2023

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

Q2. Write short answers of any eight parts from the following. (8x2=16)

(i) Name the properties used in equations: (a): 100 + 0 = 100 (b): 1000 × 1 = 1000

(ii) Separate into real and imaginary parts, if Z = i/(1+i)

(iii) Differentiate between Equal and Equivalent sets, with example.

(iv) Write the set: {x|x ∈ N ∧ 4 < x < 12}, in descriptive and tabular forms.

(v) Define semi-group.

(vi) Find values of x if |3 1 x; -1 3 4; x 1 0| = -30

(vii) If the matrices A and B are symmetric and AB = BA, show that 'AB is symmetric.

(viii) If A = [i 1+i; 1 -i], find A + $(\bar{A})^t$ [best read — partially covered]

(ix) Solve: x(x + 7) = (2x-1)(x+4) by factorization.

(x) If ω is a cube root of unity, form an equation whose roots are 2ω and $2\omega^2$

(xi) Find two consecutive numbers, whose product is 132.

(xii) Find the three cube roots of -8

Q3. Write short answers of any eight parts from the following: (8x2=16)

(i) Without finding constants write $(x^2-10x+13)/((x-1)(x^2-5x+6))$ into partial fractions.

(ii) Find vulgar fraction equivalent to recurring decimal 0.7 [best read]

(iii) Find the nth term of sequence (2/3), $(2/3)^2$, $(2/3)^3$, ...

(iv) Calculate geometric means between 4 and 16.

(v) If $y = 2x/3 + 4x^2/9 + 8x^3/27 + \ldots$ and if $0 < x < 3/2$, then show that $x = 3y/(2(1+y))$

(vi) Find 12th term of H.P: 1/3, 2/9, 1/6, ...

(vii) Find the term involving x^{-2} in the expansion of [illegible in source scan — covered by watermark]

(viii) How many words can be formed from PLANE using all letters when no letter is to be repeated.

(ix) Write formula for ${}^nP_r$ and ${}^nC_r$ [best read]

(x) A die is thrown. Find the probability that dots on top are prime numbers. [best read]

(xi) Expand (1 - x)^(1/2) up to 4 terms by binomial theorem [best read]

Q4. Write short answers of any eight parts from the following: (8x2=16) [heading covered by sticker — inferred from sequence]

(i) Define the word 'Trigonometry'

(ii) Find $\tan\theta$ and $\cot\theta$ for $\theta = 19\pi/3$

(iii) Show that $\sin^2(\pi/6) + \sin^2(\pi/3) + \tan^2(\pi/4) = 2$

(iv) Find the value of $\cos(\pi/12)$

(v) Prove that $\text{Sin}(180^o+\propto)\ \text{Sin}(90^o-\propto) = -\text{Sin} \propto \text{Cos} \propto$

(vi) Define the principal tangent function.

(vii) Prove that $\text{Sin}(\propto+\beta)\text{Sin}(\propto-\beta) = \cos^2\beta - \cos^2\propto$

(viii) Define the period of a Trigonometry function

(ix) Solve the right triangle ABC in which: $r = 90^o$, b = 68.4, c = 96.2

(x) Solve the triangle ABC if: $\beta = 60^o$, $r = 15^o$, $b = \sqrt{6}$

(xi) Find the area of triangle ABC for b = 21.6, c = 30.2, $\propto = 52^o40'$

(xii) Define the trigonometric equation.

(xiii) Find the solution of Cosec $\theta = 2$ which lie in the interval $[0, 2\pi]$

Q. SECTION-II

Note Attempt any three questions. Each question carries equal marks: (10x3=30)

5. (a) Find the matrix A if: [2 -1; -1 2] A = [0 -3 8; 3 3 -7] [best read — partially covered]

5. (b) For what values of "m" the roots of the equation $x^2 + 2(1 + 3m)x + 7(3 + 2m) = 0$ be equal? [best read — partially covered]

6. (a) Resolve into partial fractions: $(x^2+1)/((x-2)(x-1)^2)$ [best read — partially covered]

6. (b) Find the values of n and r when ${}^{n-1}C_{r-1} : {}^{n}C_{r} : {}^{n+1}C_{r+1} = 3 : 6 : 11$

7. (a) Sum the series up to n terms $2 + (2 + 5) + (2 + 5 + 8) + \ldots$

7. (b) Use binomial theorem to show that: $1 + 1/4 + 1\cdot3/(4\cdot8) + 1\cdot3\cdot5/(4\cdot8\cdot12) + \ldots = \sqrt{2}$

8. (a) Prove that $(\tan\theta+\text{Sec}\theta-1)/(\tan\theta-\sec\theta+1) = \tan\theta + \sec\theta$

8. (b) Prove that $\cos20^{o} + \cos100^{o} + \cos140^{o} = 0$

9. (a) The measures of sides of a triangular plot are 413, 214 and 375 meters. Find the measure of corner angles of the plot.

9. (b) Prove that: $\text{Sin}^{-1}\ 3/5 + \text{Sin}^{-1}\ 8/17 = \text{Sin}^{-1}\ 77/85$